COM111 Tutorial 5b

1. Use nodal analysis to find current io in the circuit of Fig. 1.

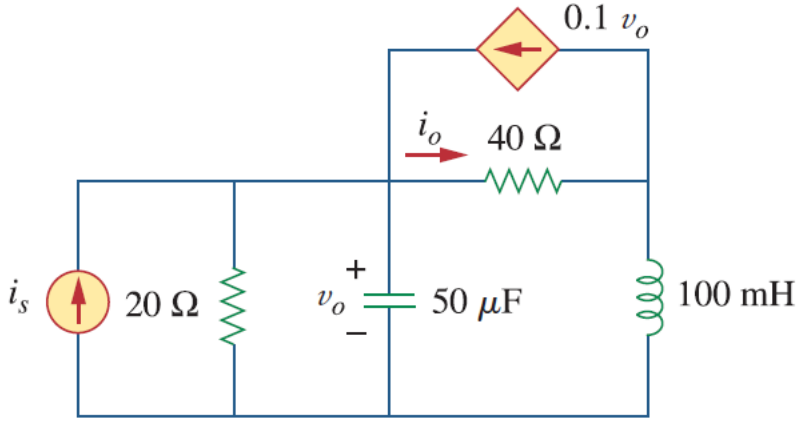


Fig. 1

2. Use mesh analysis to find vo in the circuit of Fig. 2, where v_{s1} = 120cos(100t+90°) V and v_{s2} = 80cos(100t) V.

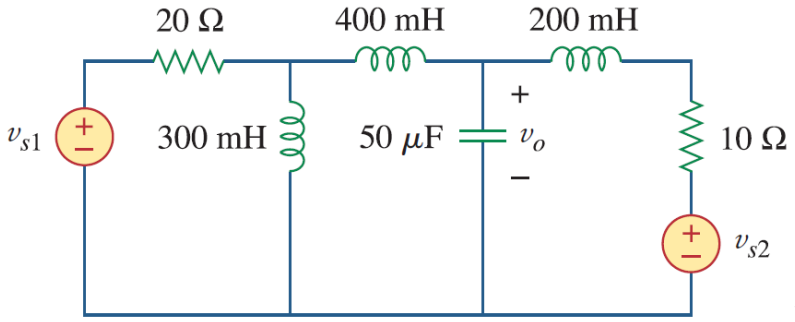


Fig. 2

3. Use superposition principle to solve for vx in Fig. 3, where v_s = 50sin(2t) V and i_s = 12cos(6t+10°) A.

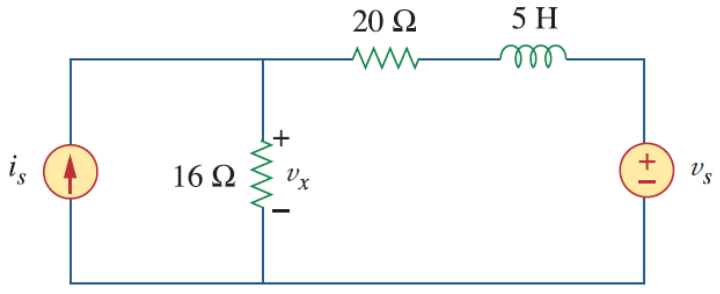


Fig. 3

4. Use the method of source transformation to find Ix in the circuit of Fig. 4.

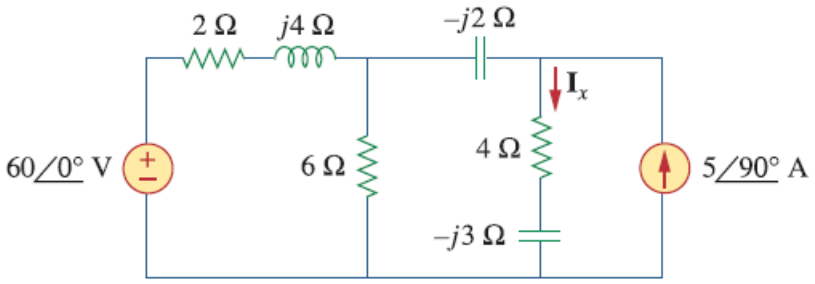


Fig. 4

5. Using Thevenin's theorem, find vo in the circuit of Fig. 5.

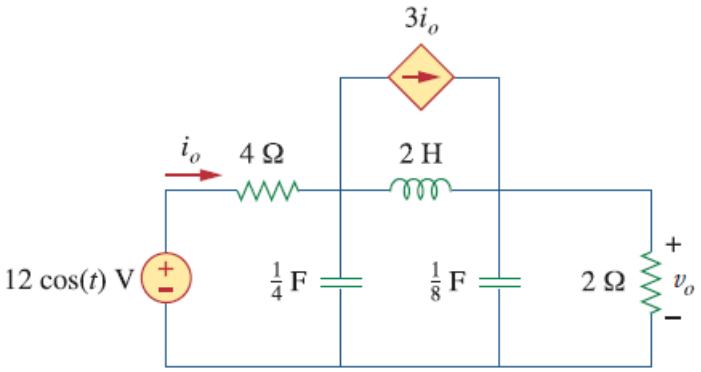


Fig. 5

6. Assuming that vs = 8cos(2t−40°) V in the circuit of Fig. 6, find the average power delivered to each of the passive elements.

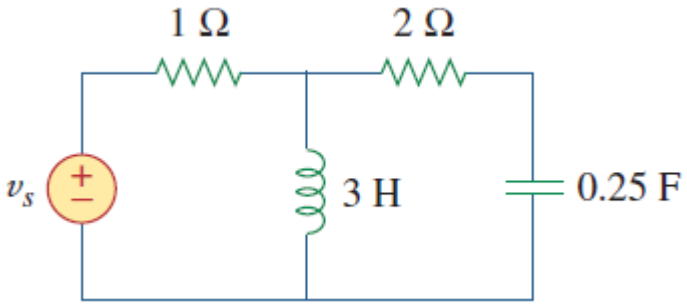


Fig. 6

7. The variable resistor R in the circuit of Fig.7 is adjusted until it absorbs the maximum average power. Find R and the maximum average power absorbed.

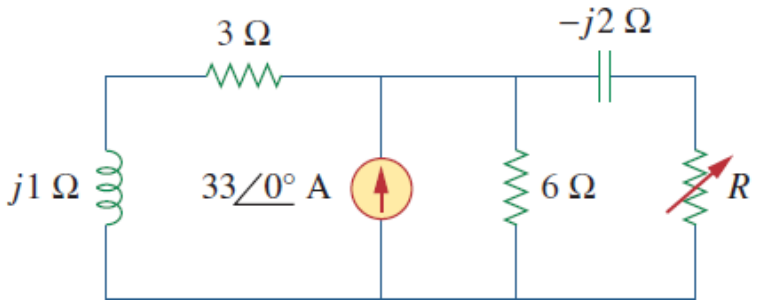


Fig. 7

8. For the power system in Fig. 8, find: (a) the average power, (b) the reactive power, (c) the power factor. Note that 220 V is an rms value.

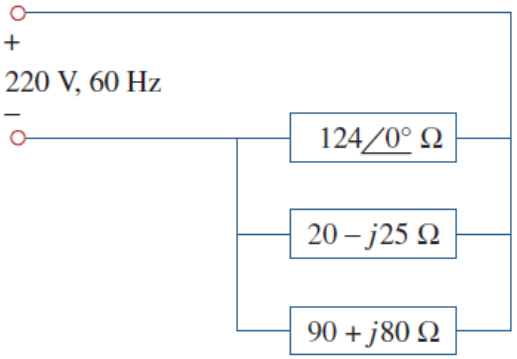


Fig. 8

9. Determine the complex power for the following cases:

   (a) $P = 269$ W, $Q = 150$ VAR (capacitive)

   (b) $Q = 2000$ VAR, pf $= 0.9$ (leading)

   (c) $S = 600$ VA, $Q = 450$ VAR (inductive)

   (d) $V_{rms} = 220$ V, $P = 1$ kW, $|\mathbf{Z}| = 40\ \Omega$ (inductive)

10. Two loads connected in parallel draw a total of 2.4 kW at 0.8 pf lagging from a 120-V rms, 60-Hz line. One load absorbs 1.5 kW at a 0.707 pf lagging. Determine: (a) the pf of the second load, (b) the parallel element required to correct the pf to 0.9 lagging for the two loads.

11. In a balanced three-phase Y-Y system, the source is an abc sequence of voltages and $\mathbf{V}an = 100 \angle 20°$ V rms. The line impedance per phase is $0.6 + j1.2\ \Omega$, while the per-phase impedance of the load is $10 + j14\ \Omega$ Calculate the line currents and the load voltages.

12. In the balanced three-phase Y- system in Fig. 9, find the line current and the average power delivered to the load.

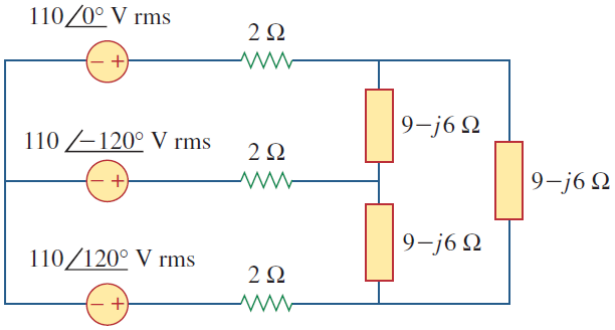


Fig. 9

13. Find the line currents Ia, Ib, and Ic in the three-phase network of Fig. 10 below. Take $Z\Delta = 12 - j15\ \Omega$, $ZY = 4 + j6\ \Omega$, and $Zl = 2\ \Omega$..

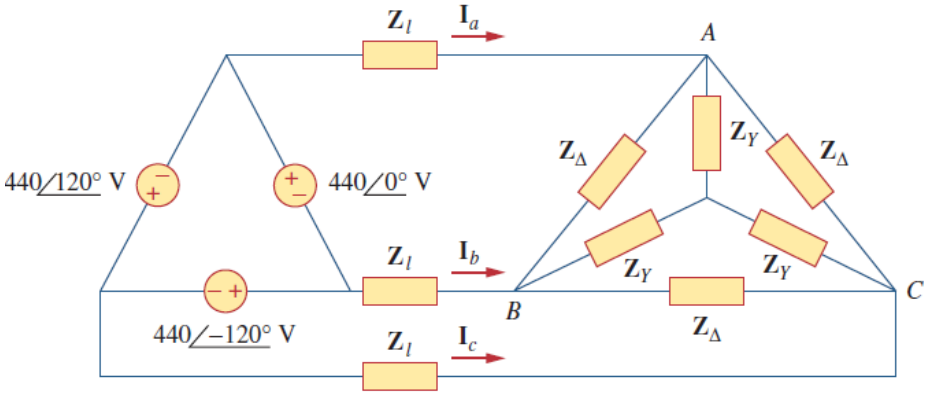


Fig. 10